

Genes and inheritance

The genes that parents pass on to their children at least partly determine their children's physical and mental characteristics. In some cases, an abnormal gene may be passed on, resulting in an inherited genetic disorder.

What are genes?

Genes are located on rodlike structures called chromosomes that are found in the nucleus of every cell in the body. Each gene occupies a specific position on a chromosome. Because genes provide instructions for making proteins, and proteins determine the structure and function of each cell in the body, it follows that genes are responsible for all the characteristics you inherit.

The full genetic instructions for each person, known as the human genome, is carried by 23 pairs of chromosomes, and consists of around 20,000–25,000 genes.

How inheritance works

At conception, the embryo receives 23 chromosomes from the mother's egg and 23 chromosomes from the father's sperm. These pair up to make a total of 46 chromosomes. Pairs 1 to 22 are identical or nearly identical; the 23rd pair consist of the sex chromosomes, which are either X or Y. Each egg and sperm contains a different combination of genes. This is because when egg and sperm cells form, chromosomes join together and randomly exchange genes between each other before the cell divides. This means that, with the exception of identical twins (see [How twins are conceived](#)), each person has unique characteristics.

How gender is determined Of the 23 pairs of chromosomes that are inherited, one pair determines gender. This pair is composed either of two X (female) chromosomes, in which case the baby will be a girl, or of one X and one Y (male) chromosome, in which case the baby will be a boy.

An egg always contains one X chromosome, while a sperm can carry an X or a Y chromosome. Whether your baby is a boy or a girl will therefore always be determined by the father. If a sperm carrying an X chromosome fertilizes the egg, the resulting embryo will be a girl. If a sperm with a Y chromosome fertilizes the egg, the resulting embryo will be a boy. In the male, both the X and Y chromosomes are active. In females, however one of the two X chromosomes is deactivated early in development of the embryo in order to prevent duplicate instructions. This could be the X chromosome from either the mother or the father.

Gene variations Each gene within a cell exists in two versions, one inherited from each parent. Often they are identical, but some paired genes occur in slightly different